AP Physics 1

Name _____

Period _____ Date _____

Practice Test – Introductory Dynamics (Version 2)

On any problems, you must show all work, include units, round properly, and box your final answer. Do the algebra first, before plugging in values.

Rewrite the following statement in the space below: On my honor, before God, I have neither given, nor received, any aid on this test.

For all questions:

- Use g = 9.8 m/s² (9.8 N/kg). Air resistance is negligible unless the question says it is not.
- Free-body diagrams: draw the forces themselves, not components, each as an arrow starting on the dot.
- Round answers to the significant figures of the given values.
- 50 minutes. Four questions, 50 points total.

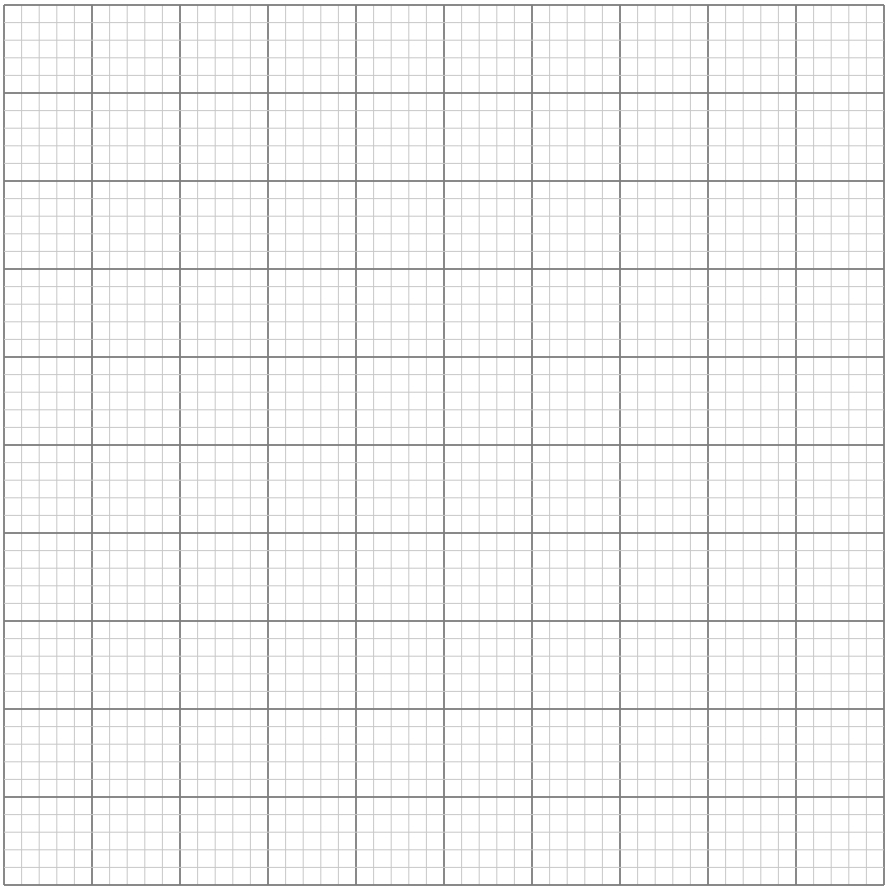
1. (14 pts) Students measure the drag force F_D on a soccer ball in a wind tunnel at several air speeds v. The drag force is modeled by $F_D = Kv^2$, where K is a constant.

Speed v (m/s)	**Drag force F_D (N)**	
10.0	0.619	
15.0	1.41	
20.0	2.58	
25.0	3.71	
30.0	5.79	
35.0	7.72	

(a) Using $F_D = Kv^2$, state the quantity you will plot on the vertical axis and the quantity you will plot on the horizontal axis so the data form a straight line, each with its units. State what the slope equals.

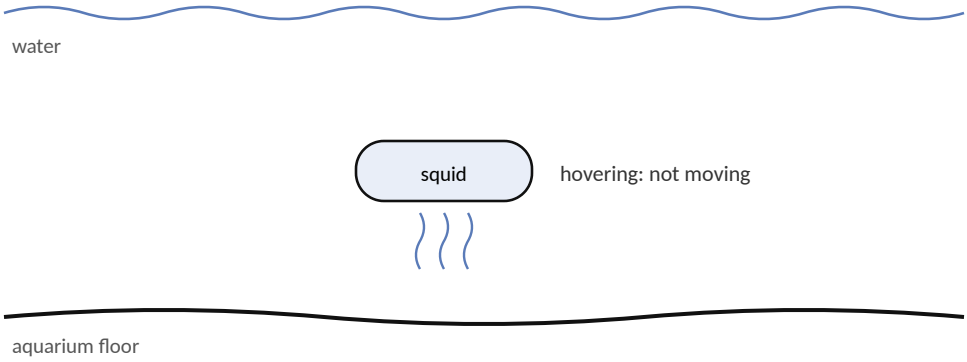
(b) Use the blank column in the table to calculate the values you will plot on the horizontal axis.

(c) Plot the data on the grid. Label each axis with the quantity and units, choose a scale, and draw the best-fit line.

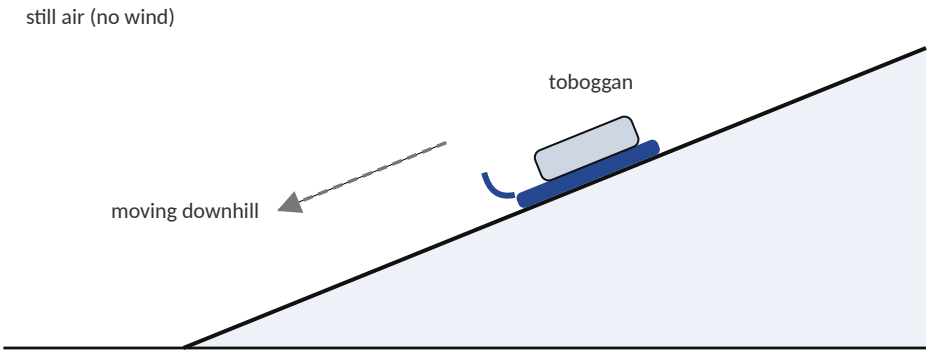


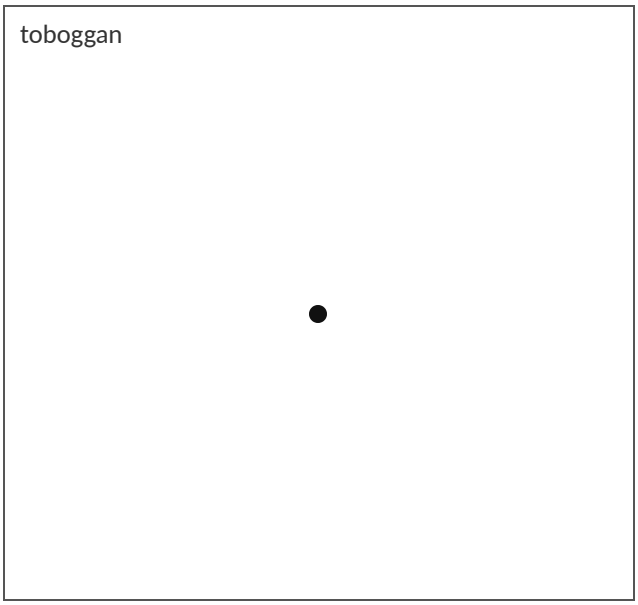
(d) Calculate the slope of your best-fit line using two well-separated points on the line. Then give K with its units.

2. (12 pts) A 45-gram squid normally rests on the floor of an aquarium. However, it can hover in the water by generating 0.080 newtons of upward thrust through jet propulsion. What is the magnitude of the buoyant force on the squid?

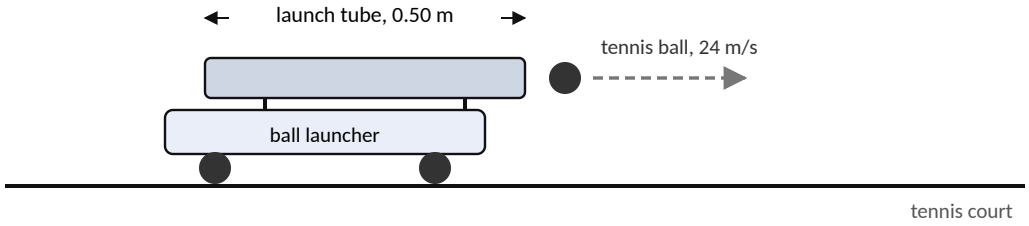


3. (12 pts) Draw a free-body diagram for a loaded toboggan sliding downhill. Friction and drag are **NOT** negligible. Label each vector you draw.





4. (12 pts) A tennis player launches a 0.057-kilogram tennis ball horizontally from a 5.0-kilogram ball launcher. The tennis ball experiences constant acceleration while inside the 0.50-meter-long launch tube, exiting at a speed of 24 meters per second. If the ball launcher were not held down and if friction with the court were negligible, what would be the magnitude of the ball launcher's acceleration?



Answer Key – Practice Test, Version 2 – Introductory Dynamics

Worked step by step: diagram, symbolic equations and algebra first, then numbers with units. $g = 9.8\ m/s^2$ does not limit significant figures.

Numerical problems: diagram first, then a = ΣF/m (equation sheet), written a = F_{net}/m. Replace F_{net} with the actual forces and their signs, swap $F_g = mg$, solve in letters, then plug in numbers with units.
Kinematics: #1 $v = v_0 + at$ · #2 $\Delta x = v_0t + \frac{1}{2}at^2$ · #3 $v^2 = v_0^2 + 2a\Delta x$. Pick the one that leaves out the variable you don't know and don't need.

1. Drag linearization (14 pts)

Step 1 **(a) Match $F_D = Kv^2$ to $y = mx + b$.**

$F_D = K \cdot (v^2) + 0$

vertical (y)	horizontal (x)	slope	intercept
F_D (N)	v^2 (m^2/s^2)	K	0

Vertical: F_D in N. Horizontal: v^2 in m^2/s^2. Slope = K.

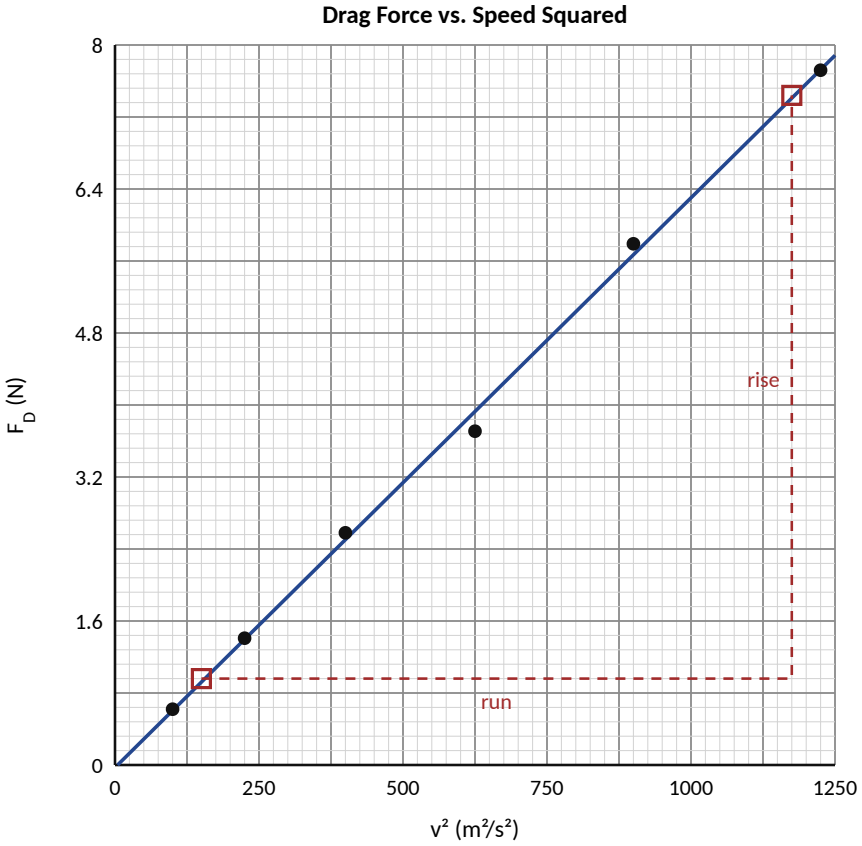
Watch out: v^2 is speed squared, in m^2/s^2. It is not an acceleration (m/s^2).

Step 2 **(b) Square each speed; the units square too.**

Example: $(10.0\ m/s)^2 = 100\ m^2/s^2$

v (m/s)	F_D (N)	v^2 (m^2/s^2)
10.0	0.619	100
15.0	1.41	225
20.0	2.58	400
25.0	3.71	625
30.0	5.79	900
35.0	7.72	1225

Step 3 **(c) Graph: labeled axes with units, a scale that uses more than half the grid, one straight line through the middle of the points.**



The best-fit line is not forced through any data point. The red squares are points read off the line, not data points.

Step 4 **(d) Slope from two well-separated points on the line, units carried.**

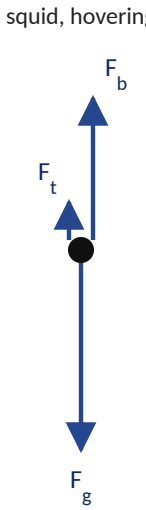
Points read off the line, away from data points: (150 m²/s², 0.96 N) and (1175 m²/s², 7.44 N)

$$\text{slope} = \frac{7.44\ \text{N} - 0.96\ \text{N}}{1175\ \text{m}^2/\text{s}^2 - 150\ \text{m}^2/\text{s}^2} = \frac{6.48\ \text{N}}{1025\ \text{m}^2/\text{s}^2} = 0.006322\ \text{N}\cdot\text{s}^2/\text{m}^2$$

$K \approx 0.00632\ \text{N}\cdot\text{s}^2/\text{m}^2\ (= \text{kg/m})$. Any value from 0.00582 to 0.00684 N·s²/m² from a reasonable line earns credit.

Units: N ÷ (m²/s²) = N·s²/m², and N = kg·m/s², so N·s²/m² = kg/m.

2. Thrust and buoyancy (12 pts)



Step 1 **Diagram of the squid while it hovers.**

F_g down (Earth). Thrust F_t up (the water it pushes down pushes back up on it). Buoyancy F_b up (the surrounding water).

No normal force: it is off the bottom. Hovering means not moving, so $a = 0$ and the two up arrows together match F_g.

Step 2 **Newton's second law, up is +, letters first.**

$a = \frac{F_{net}}{m} \rightarrow F_b + F_t - mg = ma$, with $a = 0$

$$\boxed{F_b = mg - F_t}$$

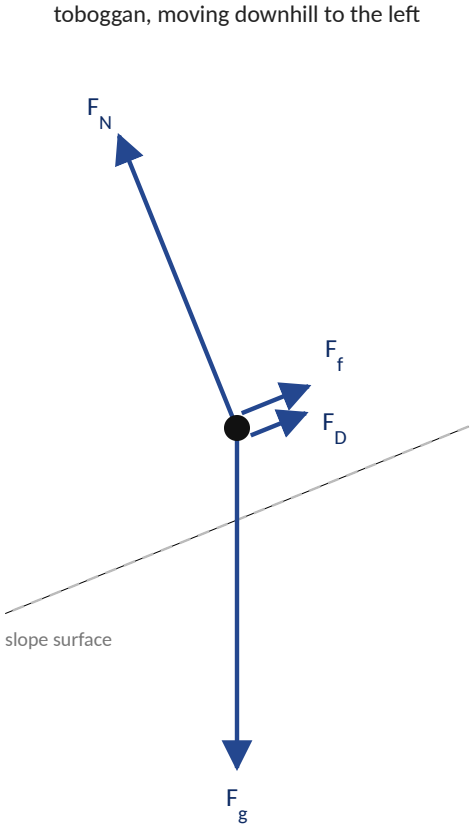
Step 3 **Numbers, units in every parenthesis.**

$m = 45\ g = 0.045\ kg$

$F_b = (0.045\ kg)(9.8\ m/s^2) - 0.080\ N = 0.441\ N - 0.080\ N = 0.361\ N \rightarrow \boxed{F_b = 0.36\ N}$

Watch out: thrust and buoyancy both point up. Writing $F_b - F_t - mg = ma$ (thrust down) flips the sign of the thrust.

3. Free-body diagram (12 pts)



Step 1 **Every force on the toboggan: exactly four.**

Earth pulls on it: F_g. The snow pushes on it two ways: F_N (perpendicular to the surface) and kinetic friction F_f (along the surface). The air pushes on it: drag F_D.

Step 2 **Directions. No angle is needed.**

F_g: straight down, vertical.

F_N: perpendicular to the slope, out of the surface, so it leans toward the left.

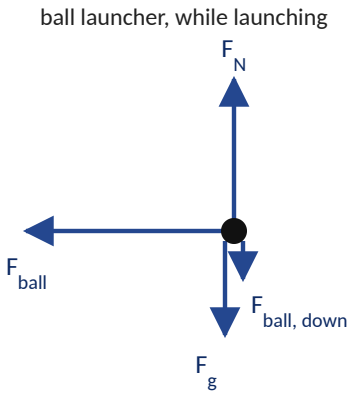
F_f: along the slope, uphill (up and to the right), opposite the sliding.

F_D: opposite the motion through the air. In still air the toboggan moves downhill through the air, so drag also points uphill, as its own arrow beside F_f.

Lengths are not graded: the motion (speeding up or not) is not stated. Every arrow starts on the dot and is labeled.

Watch out: no components of F_g drawn as extra arrows, no fifth "air resistance" arrow on top of drag, and no arrow pointing down the slope.

4. Recoil (12 pts)



Step 1 **Diagram of the ball launcher while the tennis ball is inside.**

Horizontal: only the tennis ball pushing back on it (F_{ball}). Vertical: F_N up, F_g down and the tennis ball pressing down ($F_{ball,\ down}$), all balanced, so they stay out of the horizontal equation. Not to scale.

Step 2 **Kinematics for the tennis ball: no time given, so use #3.**

$v_0 = 0$, $v = 24$ m/s, $\Delta x = 0.50$ m (the launch tube length)

$v^2 = v_0^2 + 2a\Delta x \rightarrow a_{tennis\ ball} = \dfrac{v^2}{2\Delta x} = \dfrac{(24\ \text{m/s})^2}{2(0.50\ \text{m})} =$

$576\ \text{m/s}^2 \approx$ **$5.8 \times 10^2\ \text{m/s}^2$**

Step 3 **Second law on the tennis ball, then the third law.**

$F_{net} = ma \rightarrow F = (0.057\ \text{kg})(576\ \text{m/s}^2) = 32.83\ \text{N} \approx$ **33 N**

$F_{\text{tennis ball on ball launcher}} = -F_{\text{ball launcher on tennis ball}}$: the same 33 N, backward.

Step 4 **Second law on the ball launcher.**

$a_{\text{ball launcher}} = \dfrac{F}{M} = \dfrac{32.83\ \text{N}}{5.0\ \text{kg}} = 6.5664\ \text{m/s}^2 \rightarrow$ **$6.6\ \text{m/s}^2$, magnitude (it points backward)**

Using the launch tube length as the tennis ball's distance is fine here: the ball launcher moves back only about 0.6 cm while the tennis ball is inside.

Watch out: equal and opposite forces, unequal accelerations: divide the same F by the much bigger mass.

1. Drag linearization

Suggested partial credit	pts
(a) F_D (N) vertical, v^2 (m^2/s^2) horizontal	3
(a) Slope = K	1
(b) v^2 column correct, with units	2
(c) Axes labeled with quantity and units	2
(c) Scale uses more than half the grid; points plotted	1
(c) One straight best-fit line	1
(d) Two well-separated points read off the line	2
(d) K from 0.00582 to 0.00684, with units $N \cdot s^2/m^2$ (or kg/m)	2
Total	**14**

2. Thrust and buoyancy

Suggested partial credit	pts
Diagram: F_g down, thrust and buoyancy up, no normal force	3
Hovering, so a = 0	2
$F_b + F_t - mg = ma$ with correct signs	3
Solved in letters: $F_b = mg - F_t$	1
Grams to kg; units in every parenthesis	2
$F_b = 0.36$ N	1
Total	**12**

3. Free-body diagram

Suggested partial credit	pts
F_g straight down, labeled	3
F_N perpendicular to the slope, out of the surface, labeled	3
F_f uphill along the slope, labeled	2
F_D uphill, opposite the motion, labeled, its own arrow	2
Arrows start on the dot; no components and no extra forces	2
Total	**12**

4. Recoil

Suggested partial credit	pts
#3 with $v_0 = 0$, solved for a in letters	2
$a = 5.8 \times 10^2$ m/s² for the tennis ball	2
$F = ma = 33$ N	3
Third law: same force, opposite direction, on the ball launcher	2
$a = F/M = 6.6$ m/s² (magnitude), units and sig figs	3
Total	**12**